Lecture 7 // Sept 15 // Outline of pf of Donaldson's theorem

Last Time: Let (M^4, g) be positive definite and simply connected.

Fix a $k=1$ $SU(2)$ bundle E over M.

Set $\mathcal{M} := \{ \text{SD YM connections on } E \} / \mathcal{G}$

Each splitting $L \oplus \bar{L}$ of E has a unique (up to $SU(2)$ gauge equivalence) split SDYM connection $\nabla \oplus \bar{\nabla}$ for which F_∇ is the unique harmonic representative of $c_1(L)$.

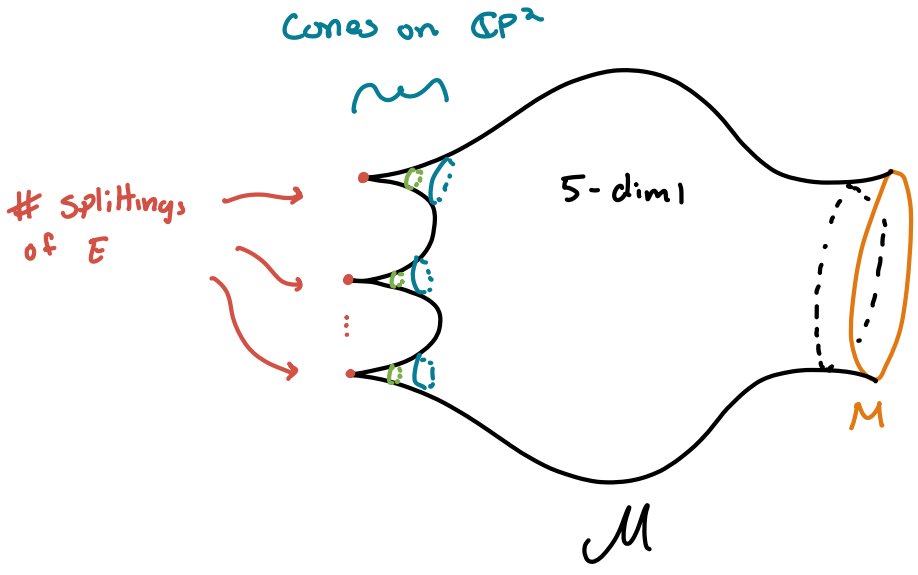
Today, we'll outline a proof of Donaldson's theorem by establishing qualitative features about $\mathcal{M}$.

Thm [Donaldson]

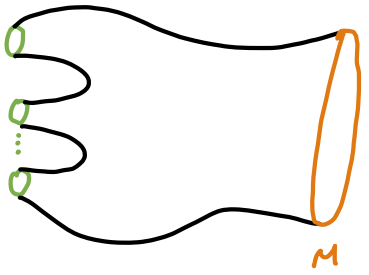
If X^4 is $\pi_1 = 1$, smooth, closed 4-mfd and Q_X is definite, then Q_X is equivalent to $n(+1)$ or $n(-1)$

Claim

1) Away from a finite number of singular points, $\mathcal{M}$ is a smooth 5-dim oriented mfd.
2) The number of these points = the # of splittings of E
3) Each point has a nbhd $\cong$ homeo to cone on $\mathbb{CP}^2$. Diffeo away from pt.
4) $\mathcal{M}$ is <u>not</u> compact, but it admits a compactification $\overline{\mathcal{M}}$ such that $\partial \overline{\mathcal{M}} \cong M$



We have a cobordism M to $\coprod \mathbb{CP}^2$ (# splittings of E) obtained by removing the singular points



(*)

It turns out that (*) implies Donaldson's theorem.

Before we discuss this, we'll briefly motivate why bordism is useful here.

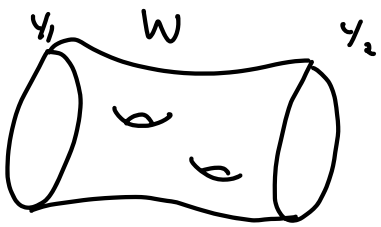
Interlude: Bordism in General

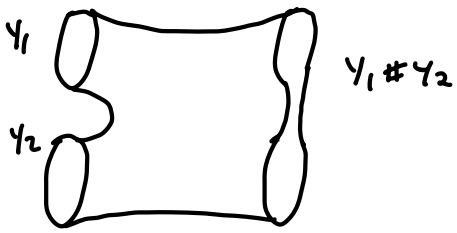
This discussion isn't strictly neccessary, but it places (*) in context.

Def The n-th bordism group is $\Omega^n = \{$ Set of all oriented smooth n-mfd. $\}/\sim$ where $Y_1 \sim Y_2$ IFF $\exists W^{n+1}$ s.t. $\partial W = -Y_1 \sqcup Y_2$

If $Y_1 \sim Y_2$, they are called bordant.

Ω^n is a grop under disjoint union and/or connect sum.





This is a generalization of the question: is $M^n \cong \partial W^{n+1}$ for some W^{n+1}?

It turns out that in one of the great alg top triumphs of the 60's, the groups Ω^n are well understood. In particular, Ω^n / torsion is known for all $n \in \mathbb{N}$. Concretely, for $n=3$, $\Omega^3 = 0$ since every 3-manifold bounds a 4-manifold. When $n=4$, $\Omega^4 \cong \mathbb{Z}$, which is determined by signature.

Def The signature of a 4-manifold is $\sigma(M) = \sigma(Q_M) = \#+$ eigenvalues $- \#-$ eigenvalues

Thm In dimension 4, signature is a complete bordism invariant.

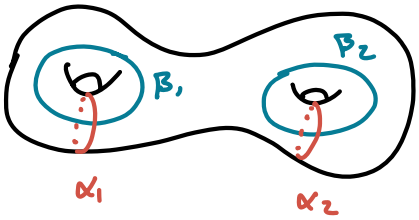
This means that two 4-manifolds are cobordant if and only if they have the same signature. In particular, any M is cobordant to either $\#_{\sigma(M)} \mathbb{CP}^2$, if $\sigma(M) \geq 0$, or $\#_{-\sigma(M)} \overline{\mathbb{CP}}^2$, if $\sigma(M) < 0$.

For our purposes, we'll only need the claim that σ is a bordism invariant.

Prop If M_1 and M_2 are bordant, then $\sigma(M_1) = \sigma(M_2)$.

To prove this proposition, we'll need the following lemma, commonly referred to as the "half lives, half dies" theorem.

To motivate it, consider $\Sigma_g = \partial H_g$. Think of Σ_g as a stand in for M and H_g a stand in for W. The middle dimension of the (co)homology of Σ is H_1, so we look at the map induced by the inclusion $\iota: \Sigma_g \to H_g$.



$$\iota_*: H_1(\Sigma_g) \longrightarrow H_1(H_g)$$

$$\ker \iota_* \cong \text{Span}\{\alpha_1, \alpha_2\}$$

$$\text{im}\, \iota_* \cong \text{Span}\{\iota_*[B_1], \iota_*[B_2]\}$$

Lemma Half lives, half dies

For W a smooth oriented $2n+1$ dimensional manifold with boundary $\partial W \cong M$, $\ker(\iota_*) \subset H_n(M;\mathbb{R})$ is a half-dimensional Lagrangian subspace with respect to the intersection form $Q: H^n(M;\mathbb{R}) \times H^n(M;\mathbb{R}) \to \mathbb{Z}$, where $\iota_*: H_n(M;\mathbb{R}) \to H_n(W;\mathbb{R})$ is the map on homology induced by the inclusion $\iota: M \cong \partial W \hookrightarrow W$

Proof of Lemma

First recall that by Poincaré duality, $H_n(M;\mathbb{R}) \cong H^n(M;\mathbb{R})$.

Further, from linear algebra, if $T: A \to B$ is a linear map,

the dual map $T^*: B^* \to A^*$ is such that $\dim \operatorname{im} T = \dim \operatorname{im} T^*$:

$$\begin{aligned}\dim \operatorname{im} T &= \dim(A) - \dim(\ker T) \\ &= \dim\left(A/\ker T\right) \\ &= \dim\left(\operatorname{Ann}(\ker T)\right) \quad \left(\begin{array}{l}\operatorname{Ann}(\ker T) = \text{Subsp of } A^* \\ \text{that annihilates } T\end{array}\right) \\ &= \dim \operatorname{im} T^*\end{aligned}$$

Thus the maps $\quad \iota^*: H^n(W;\mathbb{R}) \longrightarrow H^n(M;\mathbb{R})$

$$\iota_*: H_n(M;\mathbb{R}) \longrightarrow H_n(W;\mathbb{R})$$

are dual in the linear algebra sense, so $\dim \operatorname{im} \iota^* = \dim \operatorname{im} \iota_*$.

Again by Poincaré duality, $\dim \operatorname{im} \iota^* = \dim \ker \iota_*$.

Thus $\quad \dim \ker \iota_* + \dim \operatorname{im} \iota_* = \dim H_n(M;\mathbb{R})$

$$\begin{matrix}\| & & \| \\ \dim \operatorname{im} \iota^* & + & \dim \operatorname{im} \iota^*\end{matrix}$$

$$\Rightarrow \dim \operatorname{im} \iota^* = \frac{1}{2} \dim H_n(M;\mathbb{R}) = \frac{1}{2} \dim H^n(M;\mathbb{R})$$

$$\Rightarrow \dim \ker \iota_* = \frac{1}{2} \dim H_n(M;\mathbb{R})$$

Now, take $\alpha, \beta \in H^n(W;\mathbb{R})$, so $\iota^*(\alpha), \iota^*(\beta) \in H^n(M;\mathbb{R})$.

Then $\quad Q_M\left(\iota^*(\alpha), \iota^*(\beta)\right) = \int_M \iota^*(\alpha) \wedge \iota^*(\beta) = \int_M \iota^*(\alpha \wedge \beta) = \int_{\partial W} \iota_*(\alpha \wedge \beta)$

By Stokes' theorem, $\quad \int_{\partial W} \iota_*(\alpha \wedge \beta) = \int_W d(\alpha \wedge \beta) = 0$

Thus $\operatorname{im}(\iota^*)$ is Lagrangian with respect to Q_M. □

Rmk We may avoid using Stokes' theorem by translating to algebraic topology and cupping against fundamental class. Thus half lives half dies is true when M isn't smoothable.

Having proven half lives half dies, we may now prove the original proposition.

Prop If M_1 and M_2 are bordant, then $\sigma(M_1)=\sigma(M_2)$.

Pf 1) By setting $M=-M_1 \# M_2$, we know that $M \cong \partial(W^5)$.

So it is enough to show that $\sigma(M)=0$.

2) Consider the long exact sequence in homology and the Poincaré-Lefshetz isomorphisms. By naturality, the following diagram commutes:

$$\begin{array}{ccccccc} \cdots \to & H^2(W;\mathbb{R}) & \xrightarrow{\iota^*} & H^2(M;\mathbb{R}) & \to & H^3(W,M;\mathbb{R}) & \to \cdots \\ & \downarrow PD \cong & & \downarrow PD \cong & & \downarrow PD \cong & \\ \cdots \to & H_3(W,M;\mathbb{R}) & \to & H_2(M;\mathbb{R}) & \xrightarrow{\iota_*} & H_2(W;\mathbb{R}) & \to \cdots \end{array}$$

Note that $\operatorname{im}\iota^* \cong \ker \iota_*$ under PD

3) By Half lives, half dies, $\begin{cases} \dim \operatorname{im}\iota^* = \frac{1}{2}\dim H^2(M;\mathbb{R}) \\ \dim \ker \iota_* = \frac{1}{2}\dim H_2(M;\mathbb{R}) \end{cases}$

and Q_M vanishes on $\operatorname{im}\iota^*$ (eg, $\operatorname{im}\iota^* \subset H^2(M;\mathbb{R})$ is Lagrangian)

Thus there is a half-dim subspace on which intersection form vanishes.

4) This implies $\sigma(M)=0$:

Consider the diagonalization of Q_M over $\mathbb{R}$: $Q_M \cong \left[\begin{array}{c|c} +I & 0 \\ \hline 0 & -I \end{array}\right]$.

$$\left[\begin{array}{c|c} +I & \\ \hline & -I \end{array}\right] V = V_+ \oplus V_-$$

This splits V into $V_+ \oplus V_-$, where Q_M is $\pm$-def on $V_\pm$

Let $a=\dim V_+$ and $b=\dim V_-$. Then $a+b=\dim V_+ + \dim V_- = \dim V = 2k$.

Let $U = \operatorname{im}\iota^*$. Since Q_M vanishes on U, $U \cap V_+ = \{0\}$ and $U \cap V_- = \{0\}$.

Thus $\dim(U+V_+) = \dim(U) + \dim(V_+) = k+a \leq 2k$

$\dim(U+V_-) = \dim(U) + \dim(V_-) = k+b \leq 2k$

$\Rightarrow$ $a \leq k$, $b \leq k$. Since $a+b=2k$, $a=b=k \Rightarrow \sigma(M)=a-b=0$. $\square$

Back to Donaldson

Let's return to Donaldson and take stock of what we know:

Thm [Donaldson]

If X^4 is $\pi_1 = 1$, smooth, closed 4-mfd and Q_X is definite,

then Q_X is equivalent to $n(+1)$ or $n(-1)$

The previous discussion of bordism has given us the following three facts, from which Donaldson's theorem will follow.

(i) $\sigma(M) = rk(Q_M)$ (Since M is positive definite)

(ii) $\sigma(M) = \#$ splittings of E

Since 1) σ bordism invariant (Interlude)

2) M bordant to $\bigsqcup_{\#\text{splittings of } E} \mathbb{CP}^2$ $(*)$

(iii) $\#$ splittings of $E = \frac{1}{2} \# x \in H^2$ s.t. $\langle x, x \rangle = 1$

This follows from two claims about vector bundles over X^4:

1) $SU(2)$ bundles are determined by $c_2 \in H^4$ (true only in our setting)

2) $U(1)$ bundles are determined by $c_1 \in H^2$ (true in general)

Via the curvature form or by formal properties of Chern classes, to say $E = L \oplus \bar{L}$ implies that $c_2(E) = c_1(L) \cup c_1(\bar{L}) = -c_1(L) \cup c_1(L)$

Similarly, if $c_2(E) = -x^2$ for $x \in H^2$, the above calculation implies that $E = L \oplus \bar{L}$. Note here that different choices of x give different L's in the splitting.

Therefore, if $k = c_2(E) = 1$, we must search over $x \in H^2$ for which $x^2 = 1$.

Combining (i), (ii), and (iii), we can see that

$$\sigma(M) = \mathrm{rk}(Q_M) = \#\text{ splittings of } E = \frac{1}{2}\#\{x \in H^2 \text{ s.t. } x^2 = 1\}.$$

Heuristically, this equality says that Q_M has "many" x with $\langle x, x\rangle = 1$ where "many" is exactly twice the number of splittings.

To prove Donaldson's theorem, it remains to show that Q_M is equivalent to $n(+1)$ or $n(-1)$, i.e., $\pm n = \mathrm{rk}(Q_M) = \frac{1}{2}\#\{x \in H^2 \text{ s.t. } x^2 = 1\}$.

Lemma Let $Q : V \times V \to \mathbb{Z}$ be a positive-definite symmetric unimodular bilinear pairing. Then $\#\{x \text{ s.t. } Q(x,x) = 1\} \leq 2\,\mathrm{rk}\,Q$ with equality IFF Q is equivalent 1 over $\mathbb{Z}$

Assuming the lemma, since $\mathrm{rk}(Q_M) = \frac{1}{2}\#\{x \in H^2 \text{ s.t. } x^2 = 1\}$, where x^2 is shorthand for $Q(x,x)$, we get that

$$\#\{x \text{ s.t. } Q(x,x) = 1\} \leq 2 \cdot \frac{1}{2}\#\{x \in H^2 \text{ s.t. } Q(x,x) = 1\}$$

$$\#\{x \text{ s.t. } Q(x,x) = 1\} \leq \#\{x \in H^2 \text{ s.t. } Q(x,x) = 1\}$$

So clearly we must have equality, so the lemma implies that Q is is equivalent to 1 over $\mathbb{Z}$, thus $Q \cong \pm(n1)$.

This proves Donaldson's theorem.

We now prove the lemma.

Pf of lemma First note that if $Q = I$, there is clearly $2 \cdot \mathrm{rk}\,Q$ of such elements of H^2, since the standard basis vectors e_i have self pairing 1, as well as $-e_i$.

For a general Q, we'll apply the Gram-Schmidt process over $\mathbb{Z}$. Let x be such that $Q(x,x) = 1$. We claim that $V = \langle x\rangle \oplus \langle x\rangle^\perp$.

Indeed, since any $z \in V$ decomposes as

$$z = \underbrace{Q(x,x)x}_{\in \langle x\rangle} + \underbrace{(z - Q(x,z)x)}_{\in \langle x\rangle^\perp}$$

we know that $V = \langle x\rangle + \langle x\rangle^\perp$; since $\langle x\rangle \cap \langle x\rangle^\perp = \{0\}$,

this decomposition is orthogonal.

Thus $Q: \langle x\rangle \oplus \langle x\rangle^\perp \times \langle x\rangle \oplus \langle x\rangle^\perp \to \mathbb{Z}$, so Q splits as a block matrix $Q = \left[\begin{array}{c|c} 1 & 0 \\ \hline 0 & * \end{array}\right]$ $\begin{matrix} \} x \\ \} x^\perp \end{matrix}$

Now, notice the following:

1) Q is still positive definite on $x^\perp$

2) Any $z \in V$ such that $Q(z,z)=1$ is either $\pm x$ or lies entirely in $\langle x\rangle^\perp$: If $z = cx + w$ for $w \in x^\perp$,

$$Q(z,z) = c^2 + (w,w) \neq 1$$

3) The rank of $Q|_{\langle x\rangle^\perp}$ is $\operatorname{rank}(Q)-1$

Thus we may proceed by induction on $Q|_{\langle x\rangle^\perp}: \langle x\rangle^\perp \times \langle x\rangle^\perp \longrightarrow \mathbb{Z}$.

If $\langle x\rangle^\perp$ contains no elements for which $x^2=1$, the process concludes, and the number of elements for which $x^2=1$ is necessarily less than $2\cdot \operatorname{rk}(Q)$. Otherwise, Q can be completely diagonalized, so Q is equivalent to I. $\square$